

Nuclear Structure Investigations of Light Nuclei with the R3B Experiment

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R³B Experiment @FAIR

Total Interaction Cross Section $^{12}\text{C} + ^{12}\text{C}$

Quasi-Free Scattering $^{12}\text{C}(p,2p)^{11}\text{B}$

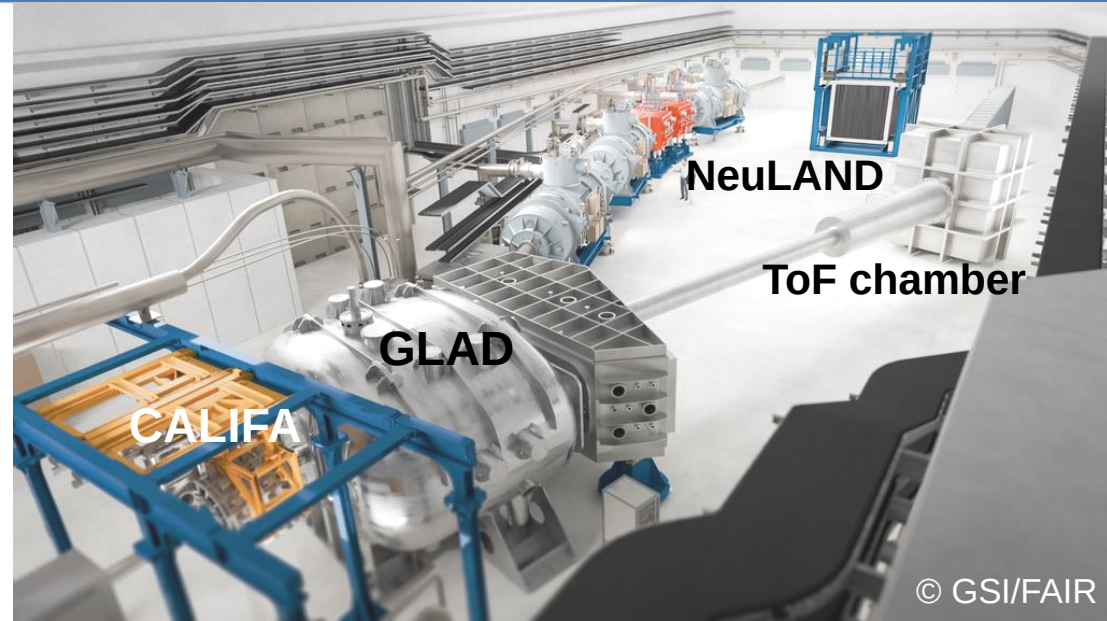
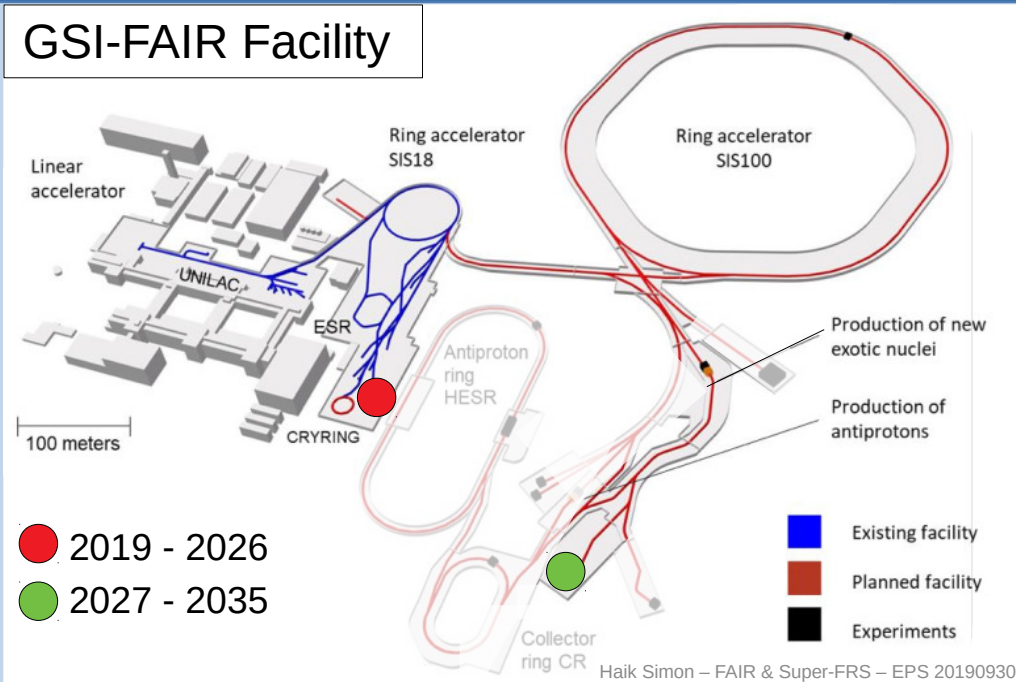
Fission Experiments @ R3B

ML for Cluster Reconstruction in CALIFA

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GSI-FAIR Facility

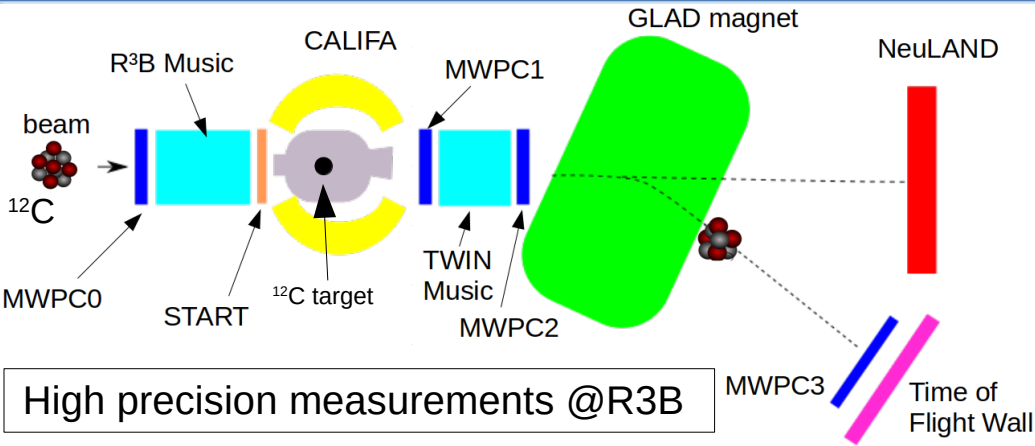


Experimental Approach:

- Inverse kinematics
- Kinematically complete measurements

Research Areas of My PhD Thesis

- Benchmark measurements for the EoS experiments
- Validation of QFS-reaction $^{12}\text{C}(p,2p)^{11}\text{B}$ and complementary fission studies
- Instrumental Development



Motivation:

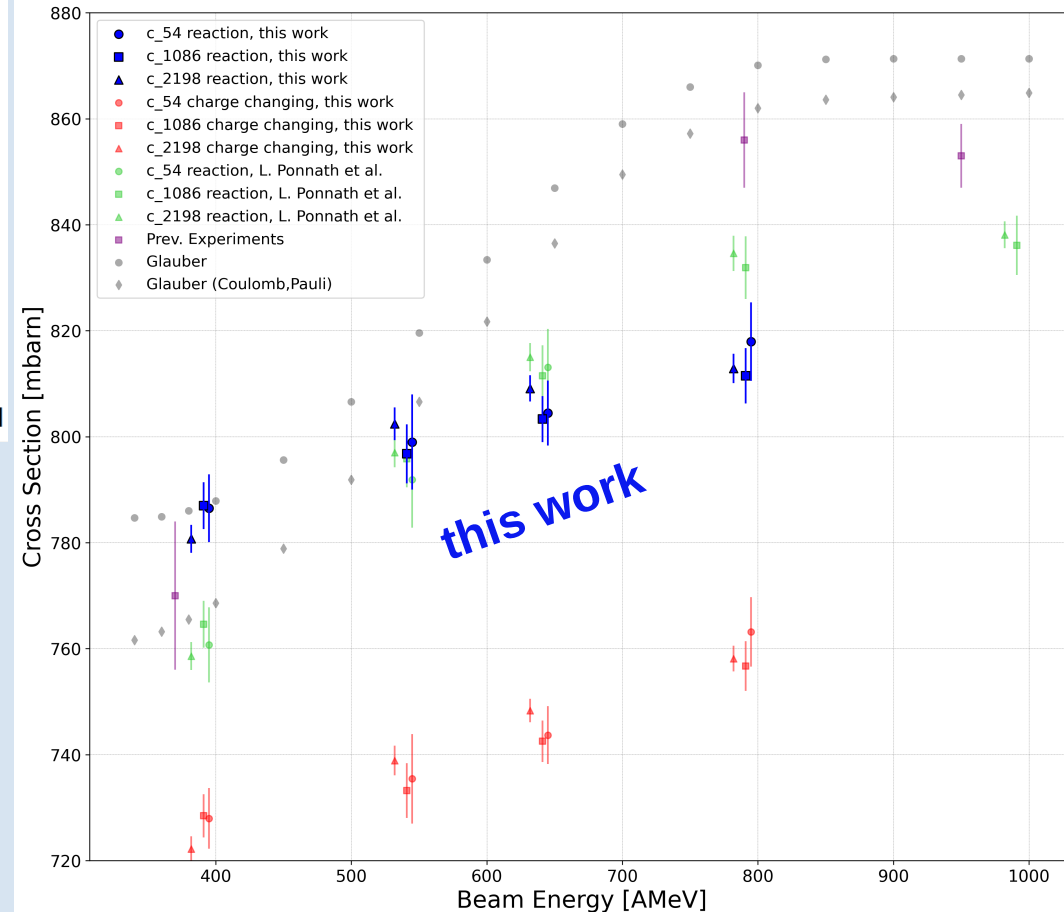
Quantify the uncertainty of the reaction model

Future physics reach:

Constrain slope parameter L of the symmetry energy of the EoS via precise $\sigma_{\Delta N}$ measurements

Measurement via transmission method:

$$N_2 = N_1 e^{-N_t \sigma}$$

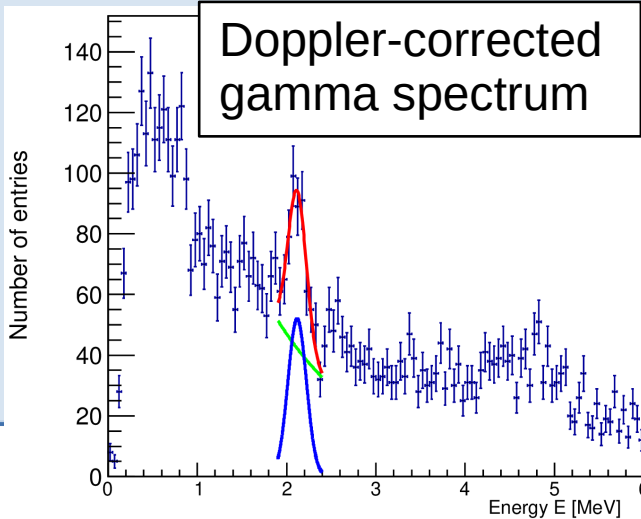
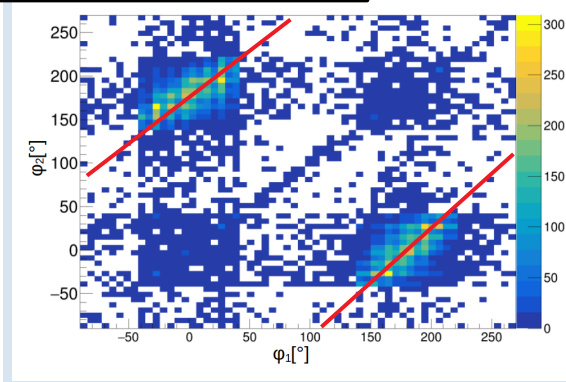
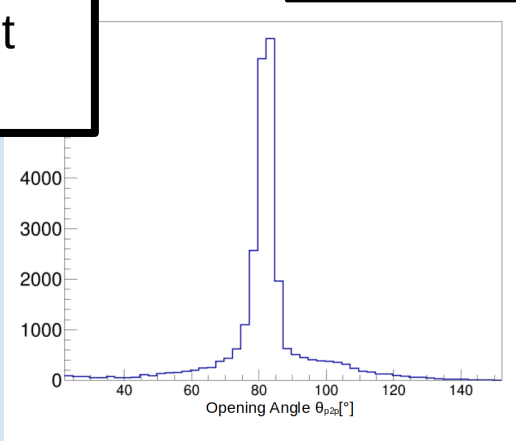
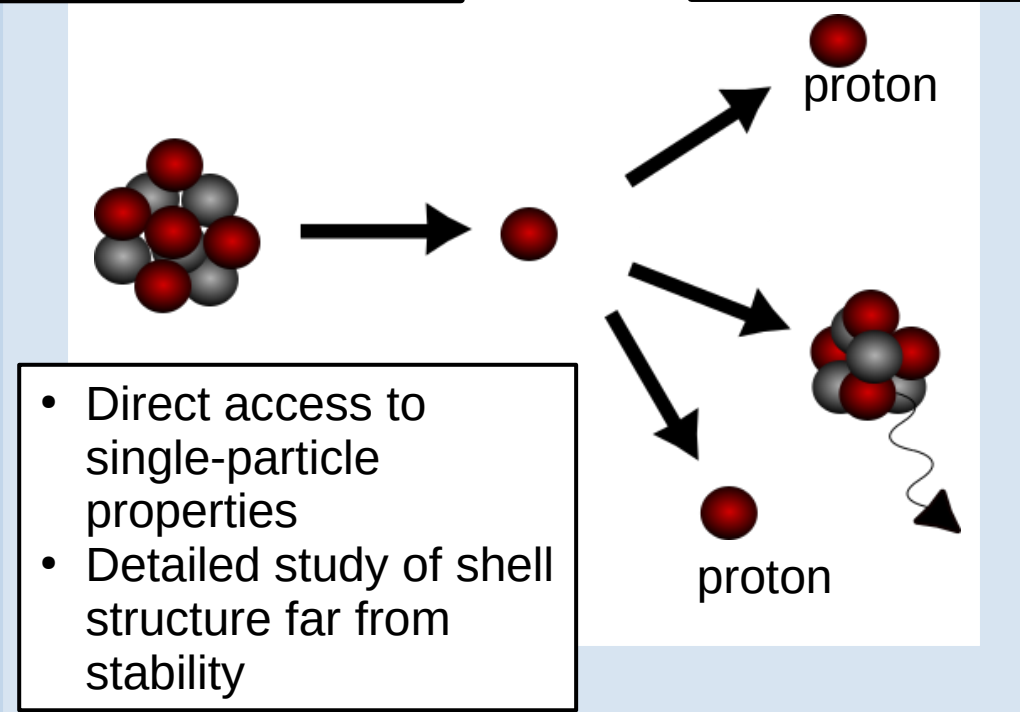


Quasi-Free Scattering $^{12}\text{C}(p,2p)^{11}\text{B}$

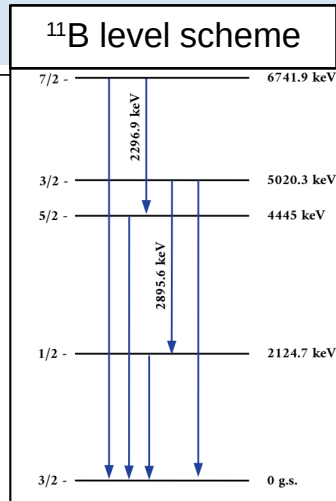
➤ ^{12}C beam
➤ proton like target

➤ 2 protons
➤ ^{11}B fragment (spectator)

two-proton correlations



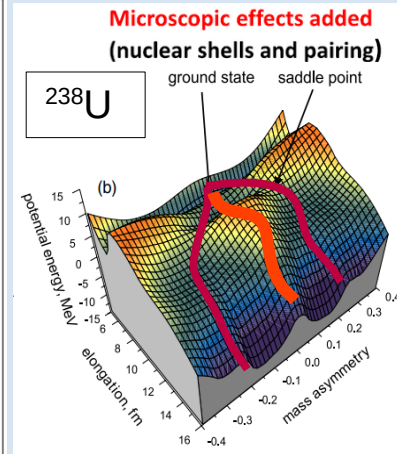
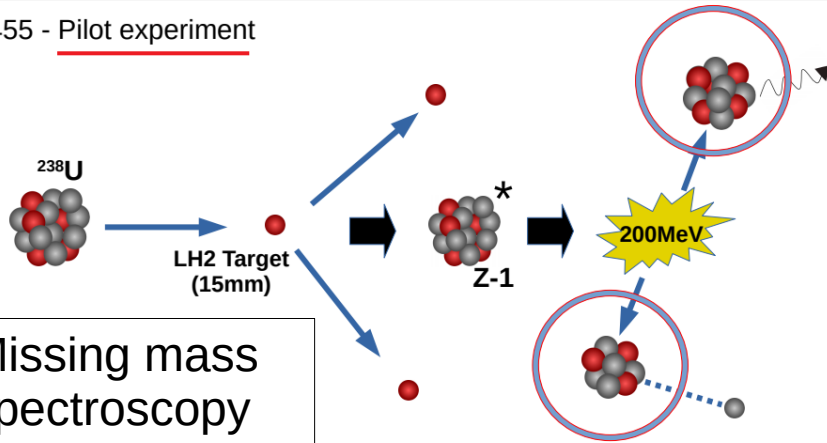
Doppler-corrected gamma spectrum



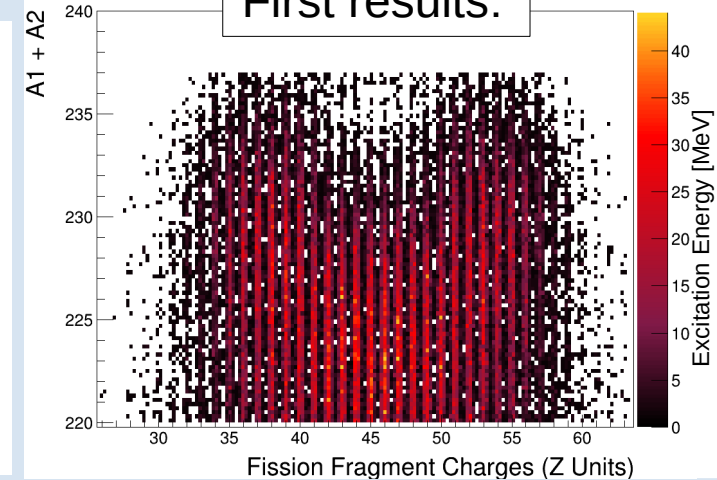
Fission via QFS: $^{238}\text{U}(p,2p\text{f})$

S455 - Pilot experiment

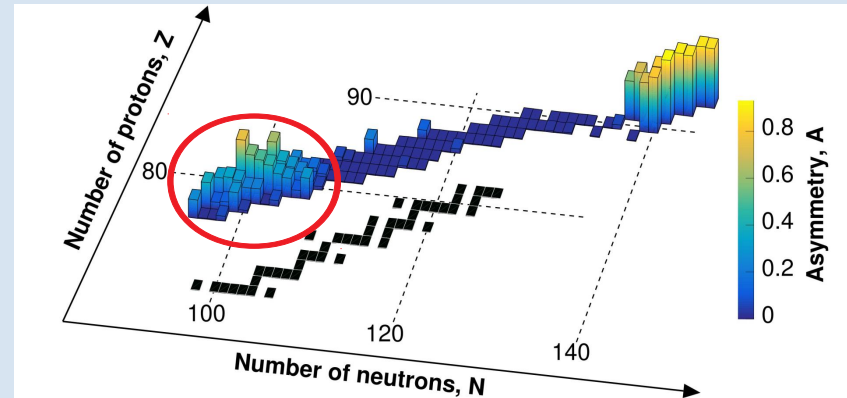
Missing mass spectroscopy



First results:

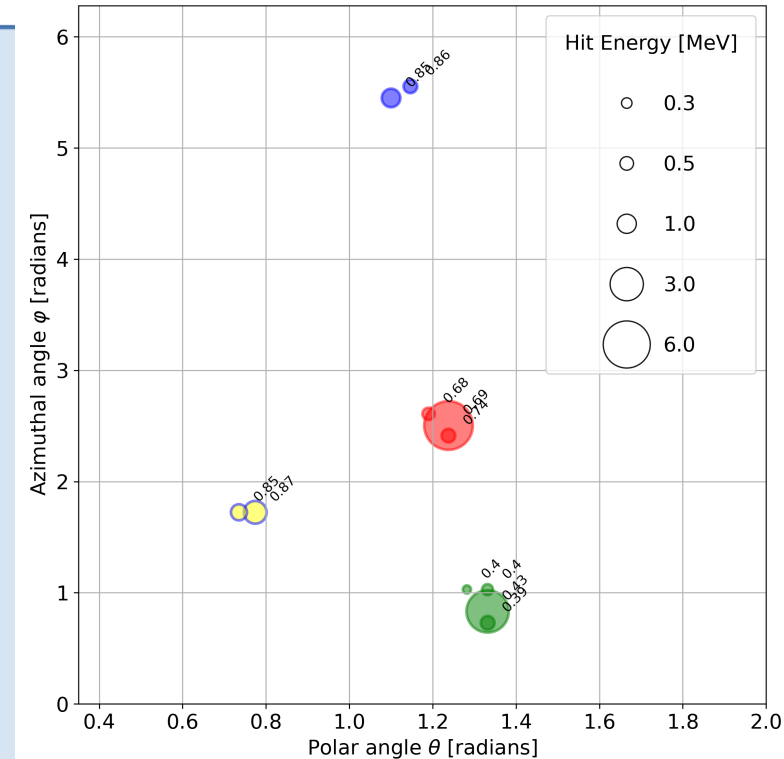
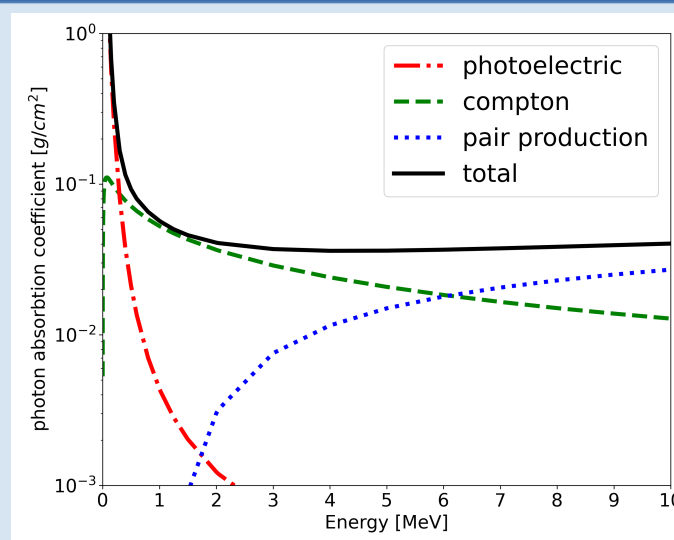
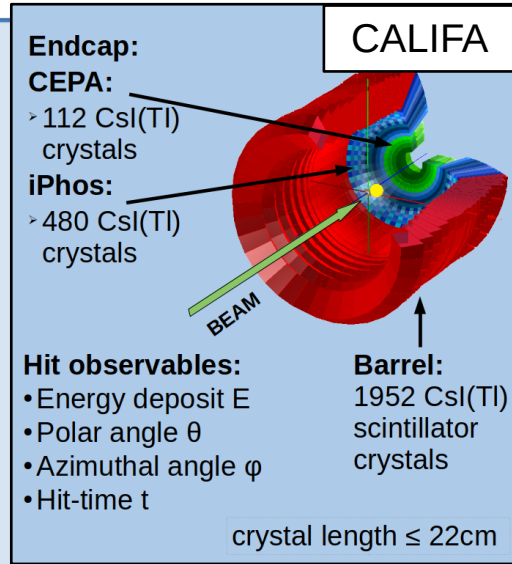


Complementary method via coulomb excitation



"An asymmetric fission island driven by shell effects in light fragments", Nature (2025)

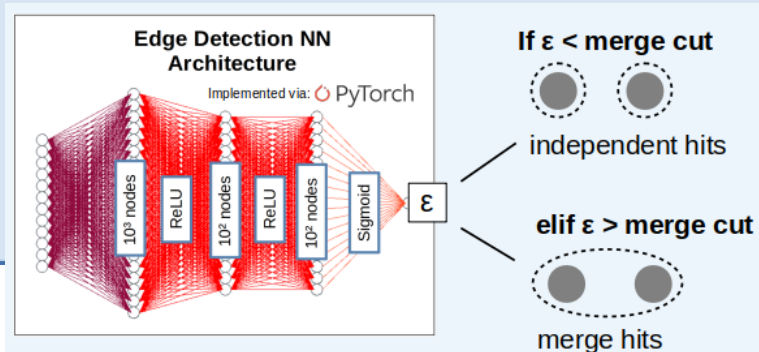
Machine Learning for the Cluster Reconstruction in the CALIFA Calorimeter at R3B



Edge Detection Neural Network

2. Pairwise hit comparison (i, j)

1. Pre-clustering



Jenegger, T. et al. (2025). ML for Cluster Reconstruction in the CALIFA Calorimeter (R3B), doi:<http://10.1016/j.nima.2025.171048>

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